

Final Exam (Summer Course)

Q1: True or False?

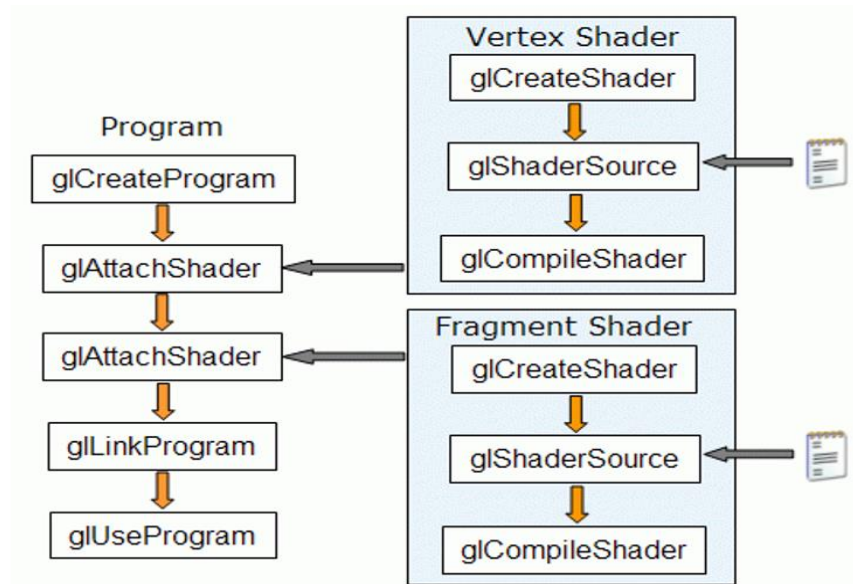
- 1- The OpenGL shading language statement "float foo = 1" is incorrect one. **True**
- 2- Reflection of a 3D object around a given axis is equivalent to its rotation by 90° around that axis. **False**
- 3- Translation and Rotation are considered to be rigid-body transformations, which move objects without any deformation or distortion. **True**
- 4- If the rotation matrix R rotates a point by the angle θ , then the inverse of R rotates the point by $-\theta$. **True**
- 5- In orthographic projection, objects that are further away from the viewer appear smaller. **False**
- 6- The term ambient means that light is perfectly reflected in a mirror-like way from the light source to the viewer. **False**
- 7- The OpenGL Shading Language (GLSL) enables programmers to replace the OpenGL fixed-function with vertex and fragment shaders. **True**
- 8- OpenGL runs only on Microsoft operating system. **False**

Q2: Fill

- 1- Given four homogeneous points $P_0 = (3, 4, 2, 95)$, $P_1 = (6, 8, 4, 1)$, $P_2 = (30, 40, 5, 5)$, and $P_3 = (12, 16, 8, 2)$. All of them except the point**P2**.....represent the same 3D point.
- 2- Given $u = (1, -1, 1)$, $v = (1, -3, 2)$ the dot product $u \cdot v$ equals.....**6**.....
- 3- Given $u = (1, -1, 1)$, $v = (1, -3, 2)$ the cross product $u \times v$ equals.....**(1, -1, -2)**.....
- 4- Given $n = (1, 1, 1)$ the normal of a plane and a point $p = (1, 1, 1)$ in the plane, then the equation of the plane is..... **$x + y + z = 3$**
- 5- if we use the drawing mode `GL_TRIANGLE_STRIP` with six vertices between `glBegin(GL_TRIANGLE_STRIP)` and `glEnd()`, the number of drawn triangles is.....**4**.....
- 6- ...**uniform**..... parameters are used to send data to vertex and fragment shader.
- 7- The OpenGL command to reflect a 2-D object around the y-axis is **`glScalef(-1.0, 1.0, 1.0)`**
- 8- ...**varying**..... parameters are passed from vertex shader to fragment shader.
- 9- The vertex shader is invoked.....**6**..... times when two triangles are sent as input for drawing by the hardware, each triangle has 2000 fragments.
- 10- The geometry shader is invoked.....**2**..... times when two triangles are sent as input for drawing by the hardware, each triangle has 2000 fragments.

Q3: Draw a block diagram illustrating the process of compiling and linking the vertex and fragment shaders:

Process Overview



Q4: Write down the transformation matrix that corresponds to the following OpenGL code segment.

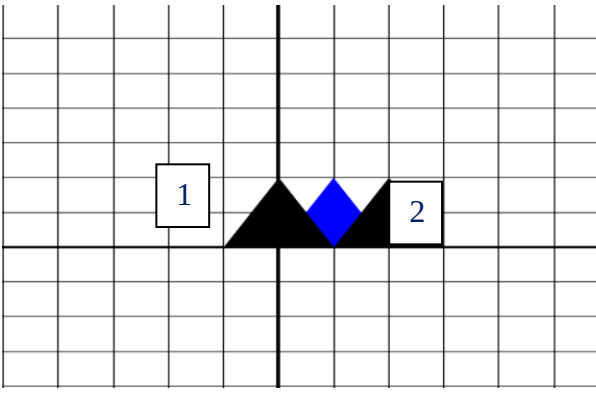
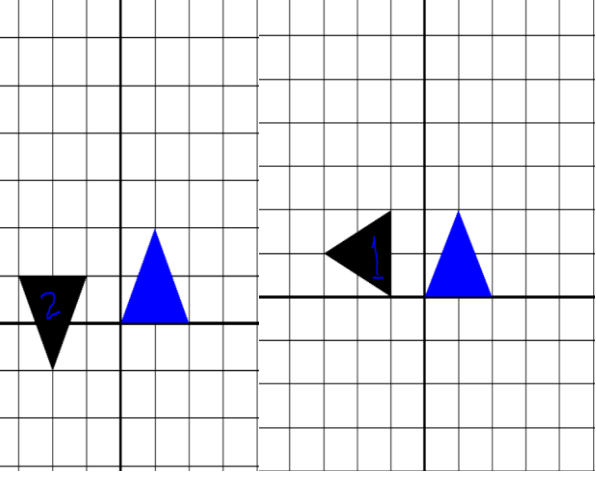
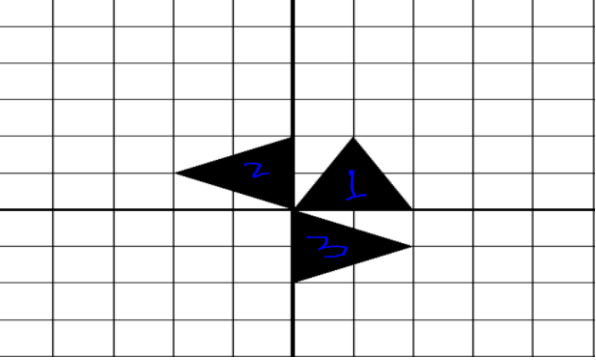
```

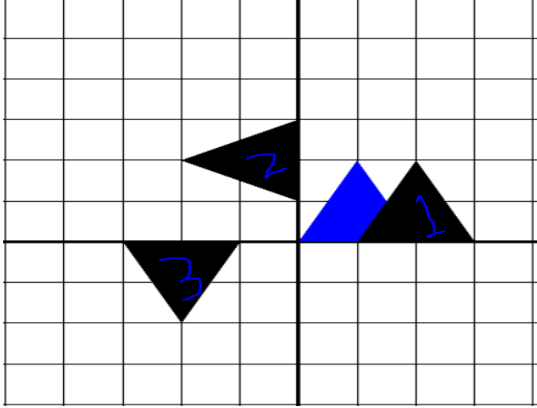
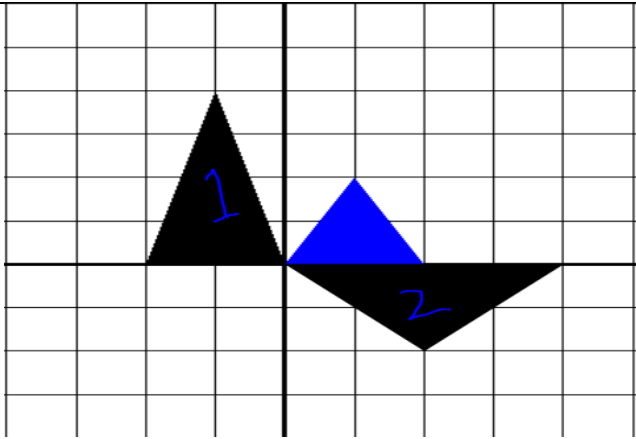
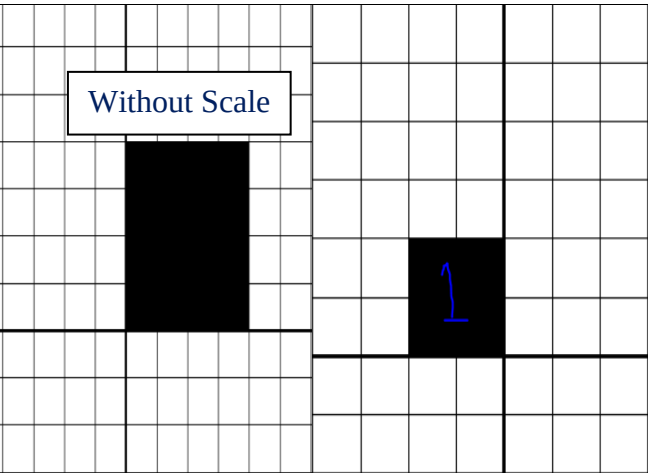
glRotatef(90.0,0,0,1.0);
glScalef(-2.0, 2.0,-1.0);
glTranslatef(2.0, 1.0, 2.0);
    
```

$$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -2 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -2 & 0 & -2 \\ -2 & 0 & 0 & -4 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Q5: What would the following codes draw Independently? Assume that the function `drawTriangle1()` draws a triangle in the plane $z = 0$, at $(0,0)$, $(2,0)$, and $(1,2)$.

<code>glMatrix Mode(GL MODEL VIEW); glLoadIdentity();</code>	
(a) <pre>glPushMatrix(); glTranslatef(-1.0, 0.0, 0.0); drawTriangle1(); → (1) glTranslatef(2.0, 0.0, 0.0); drawTriangle1() → (2) glPopMatrix();</pre>	
(b) <pre>glPushMatrix(); glTranslatef(-1.0, 0.0, 0.0); glRotatef(90, 0, 0, 1); drawTriangle1(); → (1) glTranslatef(1.0, 0.0, 0.0); glRotatef(90, 0, 0, 1); draw Triangle1(); → (2) glPopMatrix();</pre>	
(c) <pre>glPushMatrix(); for (int i = 0; i<3; i++) { glRotatef(90 * i, 0, 0, 1); drawTriangle1(); } glPopMatrix();</pre>	

(d) <pre> glPushMatrix(); for (int i = 0; i<3; i++) { glPushMatrix(); glRotatef(90 * i, 0, 0, 1); glTranslatef(1.0, 0.0, 0.0); drawTriangle1(); glPopMatrix(); } glPopMatrix(); </pre>	
(e) <pre> glPushMatrix(); glScalef(-1,2, 1.0); drawTriangle1(); → (1) glScalef(-2, -0.5, 1.0); drawTriangle1(); → (2) glScalef(0.0, -4.0, 1.0); drawTriangle1(); → (3) <u>The object disappeared</u> glPopMatrix(); </pre>	
(f) <pre> glPushMatrix(); glScalef(-0.5, 0.5, 1.0); glBegin(GL QUADS); → (1) glVertex2f(0.0, 0.0); glVertex2f(0.0, 4.0); glVertex2f(4.0, 4.0); glVertex2f(4.0, 0.0); glEnd(); glPopMatrix(); </pre>	

Q6: The figure below contains two light sources and two points in 2D. The left light source emits red diffuse light = (1.0, 0.2, 0.2) and its spotlight effect=1. The right light emits blue diffuse light = (0.2, 0.2, 1.0), spot cutoff = 40, and spot exponent=2.0. The normal vectors are shown in the figure. Assume the constant attenuation=0, linear attenuation =0.0, quadratic attenuation=1.0, the material ambient = (0, 0, 0), the material emission=(0.1,0.1,0.1), the diffuse components of the material = (0.5,0.5,0.5), and the specular components of the material = (0, 0, 0) for the two points. Using the Phong reflection model, answer the following questions:

a) The normalized normal vector for the two labeled points.

- For (A): $N_A = (0,1)$
- For (B): $N_b = (1,-2) - (0,-4) = (1, 2) = (\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}})$

b) The diffuse intensity for the two labeled points for each light source.

- **Blue:**

- For (A): $L.D = (3,5) - (4,0) = (-1,5) = (\frac{-1}{\sqrt{26}}, \frac{5}{\sqrt{26}})$

$$\text{Then, diffuse intensity} = L.D \times N_A = (\frac{-1}{\sqrt{26}}, \frac{5}{\sqrt{26}}) \times (0,1) = \frac{5}{\sqrt{26}}$$

- For (B): $L.D = (3,5) - (0,-4) = (3,9) = (\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}})$

$$\text{Then, diffuse intensity} = L.D \times N_B = (\frac{-1}{\sqrt{26}}, \frac{5}{\sqrt{26}}) \times (\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}) = \frac{7\sqrt{2}}{10}$$

- **Red:**

- For (A): $L.D = (-5,5) - (-3,3) = (-2,2) = (\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

$$\text{Then, diffuse intensity} = L.D \times N_A = (\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \times (0,1) = \frac{1}{\sqrt{2}}$$

- For (B): $L.D = (-5,5) - (-3,3) = (-2,2) = (\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

$$\text{Then, diffuse intensity} = L.D \times N_B = (\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \times (\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}) = \frac{1}{\sqrt{10}}$$

c) The attenuation factor for the two labeled points for each light source.

- **Blue:**

- For (A): $d = \sqrt{(4-3)^2 + (0-5)^2} = \sqrt{26}$

$$\text{Then, attenuation factor} = \frac{1}{k_c + k_l d + k_q d^2} = \frac{1}{0+0+(\sqrt{26})^2} = \frac{1}{26}$$

- For (B): $d = \sqrt{(0-3)^2 + (-4-5)^2} = 3\sqrt{10}$

$$\text{Then, attenuation factor} = \frac{1}{k_c + k_l d + k_q d^2} = \frac{1}{0+0+(3\sqrt{10})^2} = \frac{1}{90}$$

- **Red:**

- For (A): attenuation factor = 1
- For (B): attenuation factor = 1

d) the spotlight effect for the **blue** light for point **A**.

- $d = (0, -1)$, $v = (4, 0) - (3, 5) = (1, -5) = \left(\frac{1}{\sqrt{26}}, \frac{-5}{\sqrt{26}}\right)$
- $d \cdot v = (0, -1) \times \left(\frac{1}{\sqrt{26}}, \frac{-5}{\sqrt{26}}\right) = \frac{5}{\sqrt{26}}$
- $\theta = 11^\circ < \text{spot cutoff} = 40$
- spotlight effect = $\left(\frac{5}{\sqrt{26}}\right)^{\text{spot exponent}} = \left(\frac{5}{\sqrt{26}}\right)^2 = \frac{25}{26}$

e) the spotlight effect for the **blue** light for point **B**.

- $d = (0, -1)$, $v = (0, -4) - (3, 5) = (-3, -9) = \left(\frac{-1}{\sqrt{10}}, \frac{-3}{\sqrt{10}}\right)$
- $d \cdot v = (0, -1) \times \left(\frac{-1}{\sqrt{10}}, \frac{-3}{\sqrt{10}}\right) = \frac{3}{\sqrt{10}}$
- $\theta = 18^\circ < \text{spot cutoff} = 40$
- spotlight effect = $\left(\frac{3}{\sqrt{10}}\right)^{\text{spot exponent}} = \left(\frac{3}{\sqrt{10}}\right)^2 = \frac{9}{10}$

f) The **total** color for the point **A**.

- **Blue:**

- $R = \text{diffuse_intensity} * \text{light.red} * \text{material.red} = \frac{5}{\sqrt{26}} \times 0.2 \times 0.5 = \frac{\sqrt{26}}{52}$
- $G = \text{diffuse_intensity} * \text{light.green} * \text{material.green} = \frac{5}{\sqrt{26}} \times 0.2 \times 0.5 = \frac{\sqrt{26}}{52}$
- $B = \text{diffuse_intensity} * \text{light.blue} * \text{material.blue} = \frac{5}{\sqrt{26}} \times 1.0 \times 0.5 = \frac{5\sqrt{26}}{52}$

- **Red:**

- $R = \text{diffuse_intensity} * \text{light.red} * \text{material.red} = \frac{1}{\sqrt{2}} \times 1.0 \times 0.5 = \frac{\sqrt{2}}{4}$
- $G = \text{diffuse_intensity} * \text{light.green} * \text{material.green} = \frac{1}{\sqrt{2}} \times 0.2 \times 0.5 = \frac{\sqrt{2}}{20}$
- $B = \text{diffuse_intensity} * \text{light.blue} * \text{material.blue} = \frac{1}{\sqrt{2}} \times 0.2 \times 0.5 = \frac{\sqrt{2}}{20}$

- Total Color:

- **Blue:**

$$\bullet \quad A = \frac{1}{26} \times \frac{25}{26} \times \begin{pmatrix} \frac{\sqrt{26}}{52} \\ \frac{\sqrt{26}}{52} \\ \frac{5\sqrt{26}}{52} \end{pmatrix} = \begin{pmatrix} 3.62 \times 10^{-3} \\ 3.62 \times 10^{-3} \\ 0.018 \end{pmatrix}$$

- **Red:**

$$\bullet \quad A = 1 \times 1 \times \begin{pmatrix} \frac{\sqrt{2}}{4} \\ \frac{\sqrt{2}}{20} \\ \frac{\sqrt{2}}{20} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{4} \\ \frac{\sqrt{2}}{20} \\ \frac{\sqrt{2}}{20} \end{pmatrix}$$

$$\text{Then Total Color} = \begin{pmatrix} 3.62 \times 10^{-3} \\ 3.62 \times 10^{-3} \\ 0.018 \end{pmatrix} + \begin{pmatrix} \frac{\sqrt{2}}{4} \\ \frac{\sqrt{2}}{20} \\ \frac{\sqrt{2}}{20} \end{pmatrix} + \begin{pmatrix} 0.2 \\ 0.2 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 0.557 \\ 0.274 \\ 0.289 \end{pmatrix}$$

وعفى الله عما سلف